# ECE330 – Fall 2024
# Exam 1
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# 90 min

## Wednesday, October 2, 2024

**NAME** Solutions

**NetID** _____

## Instructions:

- Clearly PRINT your name in CAPITAL LETTERS.
- Clearly write your NetID.
- This exam is closed book and closed notes.
- To get credit, you must SHOW ALL your work and provide explanations where needed.
- You may write on both sides of the exam paper. If there is anything important on the reverse side, please indicate this on the front side.
- All answers must INCLUDE UNITS whenever appropriate.
- Graphing and/or 4-function calculators are permitted, but no cellphones/laptops.
- This exam booklet contains 9 pages (including this one) with 4 problems.

**Problem 1 (25 Points)**

A single-phase source is serving three loads connected in parallel:

- Load 1 draws 100 A at a lagging PF of 0.8.
- Load 2 draws 8 kW at a lagging PF of 0.7.
- Load 3 is an impedance of (5+j2) Ω. $= 5.39\angle 21.8$

The voltage across the loads is $v(t)_{load} = 283\sin(377t)$ V. $= \frac{283}{\sqrt{2}}\angle -90°$

The impedance of each of the two wires that serves this combination of loads is (0.1+j0.1) Ω.

$= (.141\angle 45) \times 2$

a) Find the total average and reactive powers provided by the source.

P = 38.3 kW

Q= 30.3 kVAR

Need $I_{tot}$

Load 1: $|I| = 100A$, $\phi_v - \phi_i = \cos^{-1}(.8) = 36.87°$

$I = 100\angle -126.87$

Load 2: $|I| = \frac{P}{PF \cdot V/\sqrt{2}} = 57.11$ $\quad \phi_v - \phi_i = \cos^{-1}(.7) = 45.57°$

$I = 57.11\angle -135.57$ $\quad \angle I = -90 - 45.54$

Load 3: $\frac{V_{load}}{\sqrt{2}} = ZI$ $\quad I = \frac{V_{load}}{\sqrt{2}Z} = 37.13\angle -111.8°$

$I_{tot} = I_1 + I_2 + I_3 = 192.3\angle -126.57°$

$V_{Line} = I_{tot} Z_{line} = 54.2\angle -81.57°$

$V_{tot} = V_{line} + V_{Load} = 253.9\angle -88.21°$

$S_{tot} = V_{tot} I_{tot}^{*} = 48815\angle 38.36°$

$P = |S|\cos(\phi) = 38275$ W

$Q = |S|\sin(\phi) = 30297$ VAR

b) Find the source voltage and current as time domain quantities. (from ↑)

$i(t)_s$= $271.95 \cos(377t - 126.57°)$ A

$v(t)_s$= $359.1 \cos(377t - 88.21°)$ V

$I_m = \sqrt{2}\, I_{RMS}$

$V_m = \sqrt{2}\, V_{RMS}$

c) What is the combined power factor over the three loads? State whether it is leading or lagging.

Combined PF = .803

Circle one: Leading / Lagging

$S_{LOAD} = V_{LOAD} I_{tot}^{*} = 384.81 \angle 36.57$

$\cos(36.57) = .803$ PF

**Problem 2 (25 Points)**

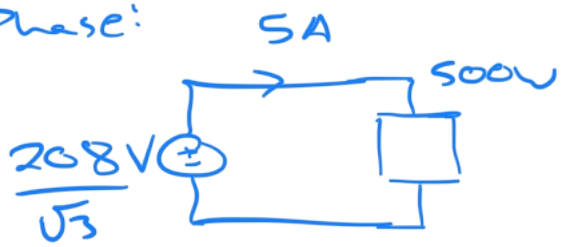
A 3-phase, 208 V (line-line), Wye-connected 60 Hz source supplies 5A (magnitude) of line current to a 1500 W, 3-phase wye-connected load at a lagging power factor.

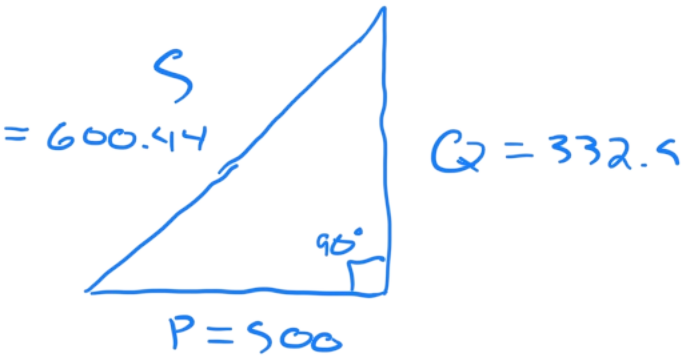
**a)** When a new wye-connected purely reactive load is connected in parallel with the 1500 W 3-phase load, the line current (magnitude) increases to 7A. What is the impedance per phase of the new wye-connected load?

**Hint: You may assume that the real power does not change when the new wye-connected load is added**.

$\bar{Z}_Y =$ 42∠90 Ω

Per Phase:




old: $|S| = VI = 600.44$

$Q = \sqrt{S^2 - P^2} = 332.5$

new: $|S|_{new} = IV = 840.6\,VA$

$Q_{tot,new} = \sqrt{S^2 - P^2} = 675.8 = Q_{old} + Q_{new}$

$Q_{new} = 343.3\,VAR \Rightarrow \bar{S} = 343.3\angle 90$

$I = \left(\frac{S}{V}\right)^* = 2.86\angle -90$

$Z = \frac{V}{I} = 42\angle 90$

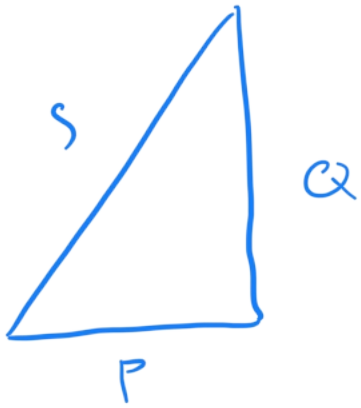
**b)** Now a delta-connected capacitor bank is added in parallel with the two loads, reducing the total line current from 7A to 4.75 A (magnitude). What is the size of the capacitors added per phase in Farads (capacitance) in the delta-connected capacitor bank?

C = 24.6 μF

$|S_{new}| = \sqrt{3}\, V_L I_L = 1711.27 \text{ VAR}$

$Q_{new} = \sqrt{\left(\frac{S_{new}}{3}\right)^2 - P^2} = 274.6$

$Q_{added} = Q_{new} - Q_{old} = -401.2$




$S = 401.3\angle -90$

$I = \left(\frac{S}{V}\right)^* = 1.93\angle 90$

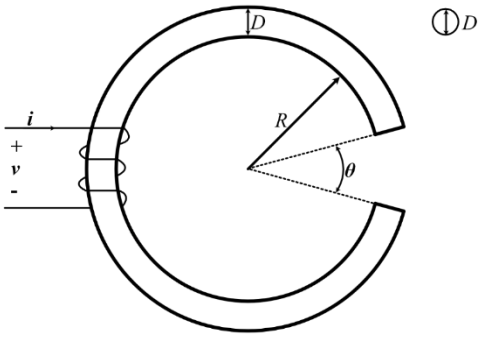
$Z_\Delta = \frac{V}{I} = 108\angle -90 = Z_C$

$Z_C = \frac{-j}{2\pi\omega C} = \frac{1}{2\pi\omega C}\angle -90^\circ \qquad \omega = 60$

$C = 2.46E\text{-}5\,F$

**Problem 3 (25 Points)**

A toroidal iron core with permeability $\mu = 1500\mu_0$ and inner radius *R*=10cm, shown below, has a coil with 100 turns wrapped around it. The cross section of the toroid is a circle, as shown, and has a diameter *D*=2 cm. There is a cut in the toroid that spans an angle $\theta$=10°. Fringing effects can be neglected.




**a)** What are the reluctances for the iron core and the air gap?

**Hint: the arc length of an arc lying on a circle with radius *r* spanning $\theta$ radians is $l = r\theta$.**

$R_{core}$ = 1,134,797 Ampere-turns / Weber

$R_{gap}$ = 48,655,343 Ampere-turns / Weber

$R = \frac{l}{\mu A}$

core: $l = 2\pi(11cm)\frac{35}{36} = .672m$

$\mu = \mu_0 \times 1500$

$A = \pi(1cm)^2 = 3.14E\text{-}4$

air gap: $l = 2\pi(11cm)\frac{1}{36} = .019m$

$\mu = \mu_0$

$A = \pi(1cm)^2 = 3.14E\text{-}4$

**b)** What is the total magnetic flux?

$\phi$ = _____

$$\Phi = \frac{Ni}{\mathcal{R}_{tot}} = \frac{100i}{\mathcal{R}_{core} + \mathcal{R}_{gap}} \text{ Wb}$$

**c)** What is the inductance of this system?

$L$ = 2.008E-4 H

$$L = \frac{N\Phi}{i} = \frac{N^2}{\mathcal{R}_{Tot}} = 2.008E\text{-}4$$

**Problem 4 (25 Points)**

Note: Parts **a)** and **b)** are unrelated.

**a)** An ideal 240V/120V, 60 Hz single-phase transformer is being used to supply power to a load drawing 1kW at 0.8 PF lagging. Note that 240V corresponds to the primary side, and 120V corresponds to the secondary side.

i. Find the complex power being supplied to the load.

$\bar{S}_L =$ $1250\angle 36.9$ VA

$|S| = \frac{P}{PF} = 1250$ VA $\quad \angle S = \cos^{-1}(.8) = 36.9$

ii. Find the load impedance.

$\bar{Z}_L =$ $11.52\angle 36.9\ \Omega$

$I = \left(\frac{S}{V}\right)^* = 10.42\angle -36.9$

$Z = \frac{V}{I} = 11.52\angle 36.9\ \Omega$

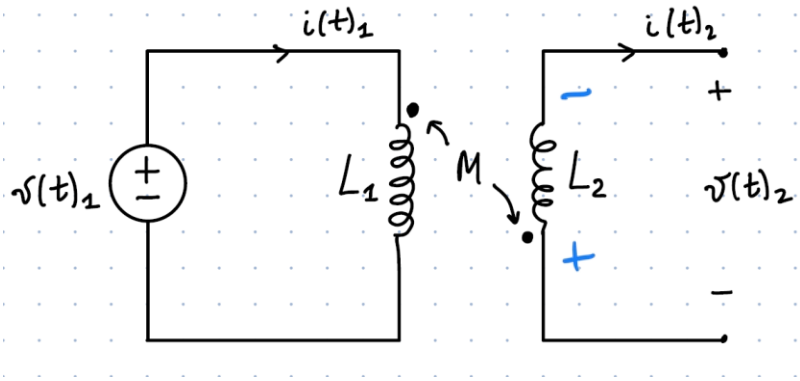
iii. Find the load impedance as seen from the primary side.

$\bar{Z}_p =$ $46.07\angle 36.9\ \Omega$

$Z_p = a^2 Z_s \quad a = \frac{N_1}{N_2} = 2$

$Z_p = 4 Z_s$

**b)** Consider two coils magnetically coupled together as shown on the diagram below.




Self -inductances of the coils are given as $L_1 = 2$ H and $L_2 = 250$ H. Mutual inductance between two coils is given as M = 5 H. Coil 2 is open-circuited. What is the voltage $v(t)_2$ when a sinusoidal voltage $v(t)_1 = 1\sqrt{2}\cos(377t)$ V is applied to coil 1?

$v(t)_2 =$ $\frac{5\sqrt{2}}{2}\cos(377t - 180°)$ V

due to dot convention

$$-V_2 = M\frac{\partial i_1}{\partial t} + L_2\frac{\partial i_2}{\partial t}$$

$$V_1 = L_1\frac{\partial i_1}{\partial t} + M\frac{\partial i_2}{\partial t}$$

$\frac{\partial i_2}{\partial t} = 0$ → open circuit

$$V_1 = L\frac{\partial i_1}{\partial t} \Rightarrow \frac{\partial i_1}{\partial t} = \frac{V_1}{L} = \frac{\sqrt{2}}{2}\cos(377t)$$

$$-V_2 = \frac{\partial i_1}{\partial t}M = -\frac{5\sqrt{2}}{2}\cos(377t)$$

$$= \frac{5\sqrt{2}}{2}\cos(377t - 180°)$$